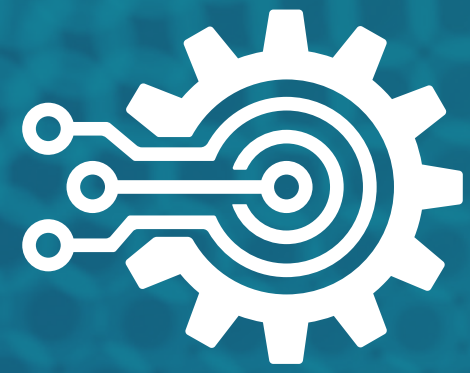
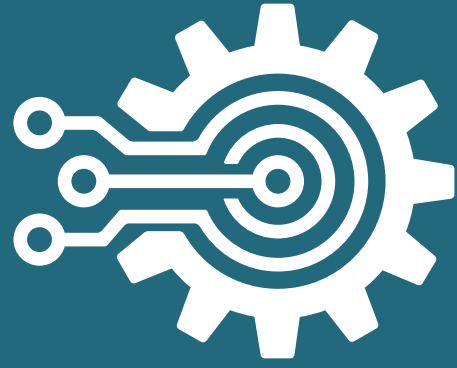




Academic Impacts of AI

Justin Southworth

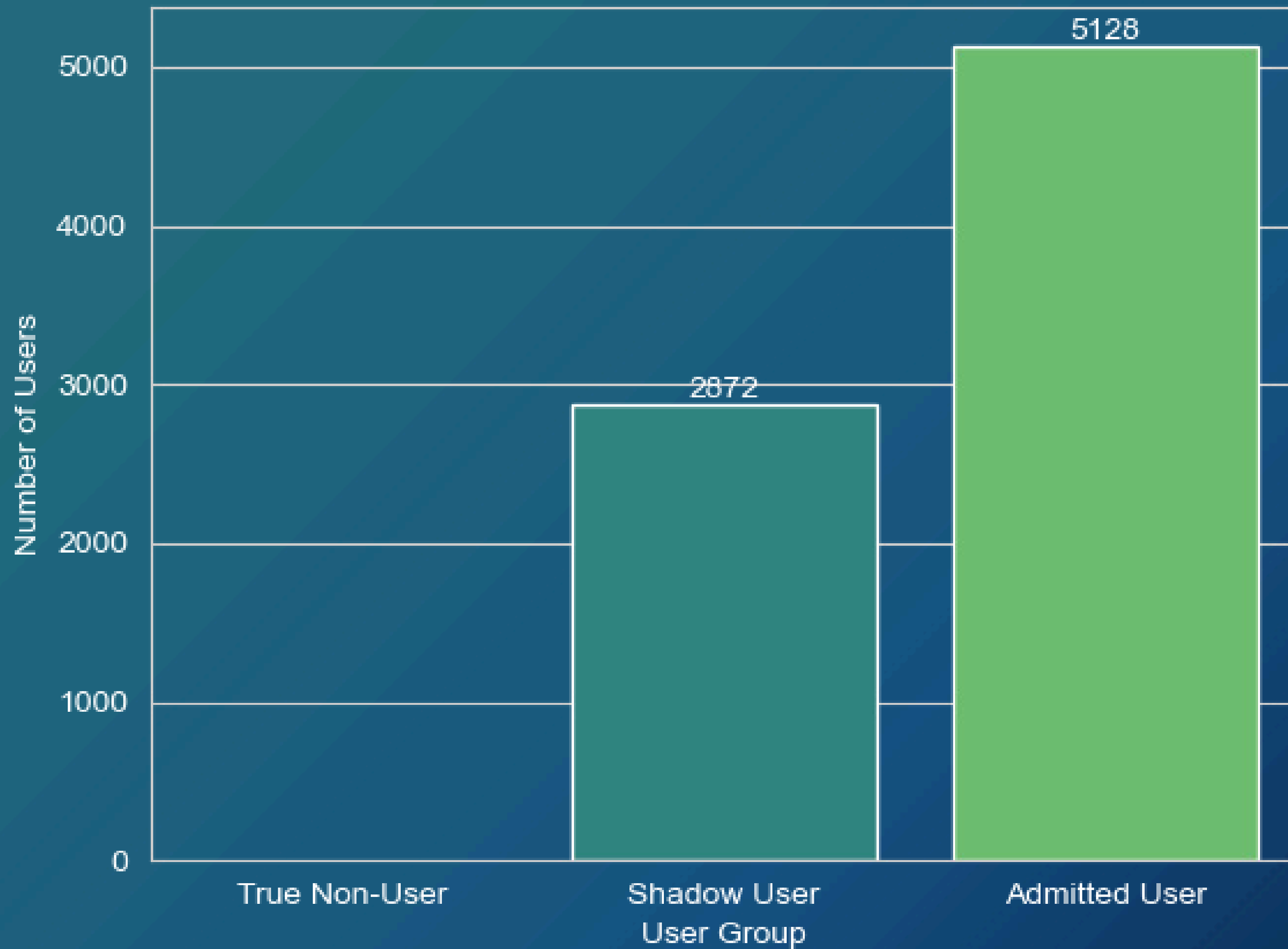
Data Source: Academic Outcomes & AI Dependency Analysis Dataset
<https://www.kaggle.com/datasets/bca5b9b9ebe556aade6ba0035cec6c627a2f8adaac329dd5544833e8633573c5/data>



Creating informed educational legislation and policies requires a clear understanding of how emerging technologies, such as Artificial Intelligence, influence student outcomes.

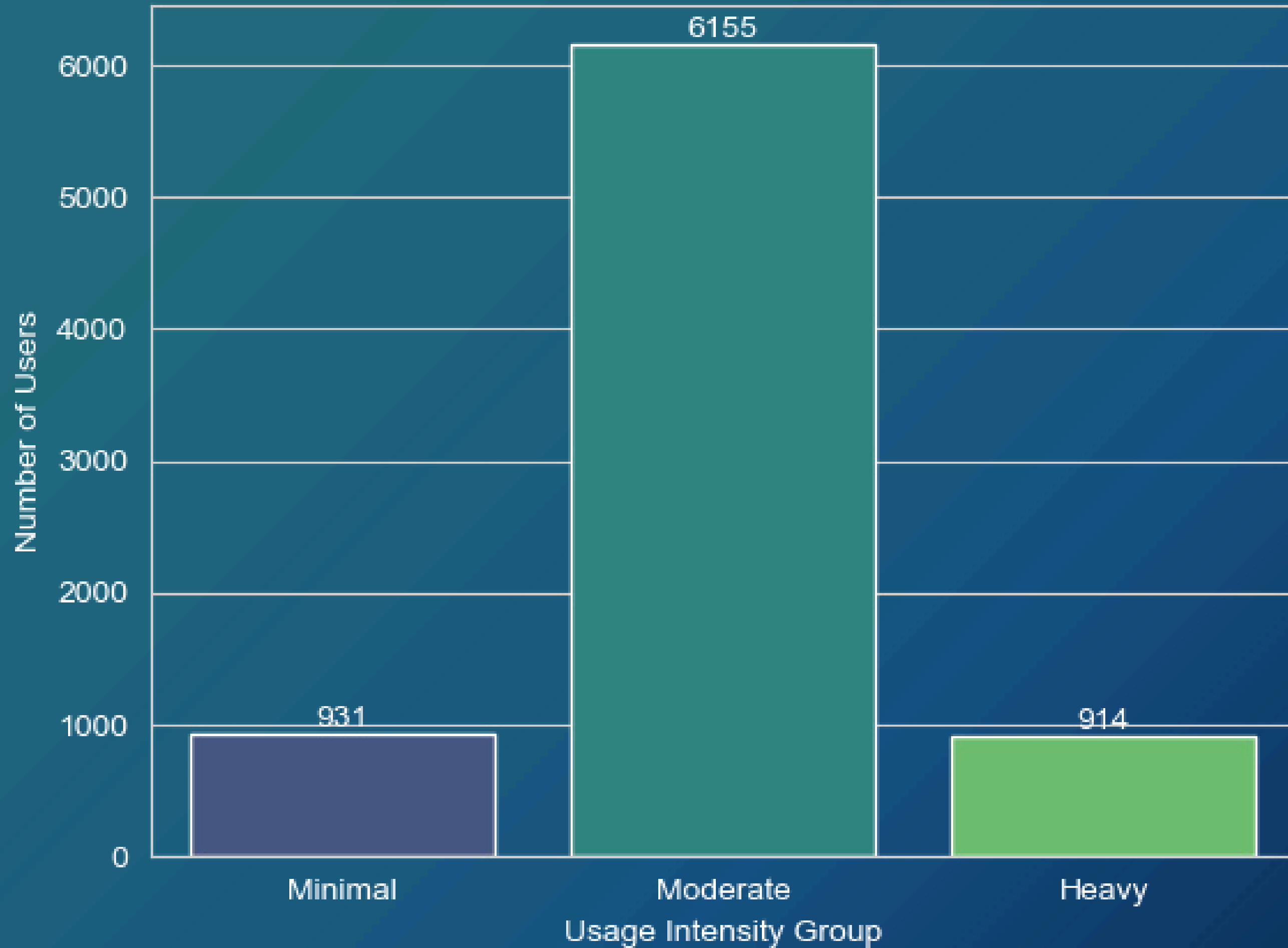
Looking at a representative population of 8,000 students over the course of a single school year, how does the use of AI—and the manner of that use—specifically affect Average Assignment Scores, Final Scores, and Improvement Rates?

Distribution of Users

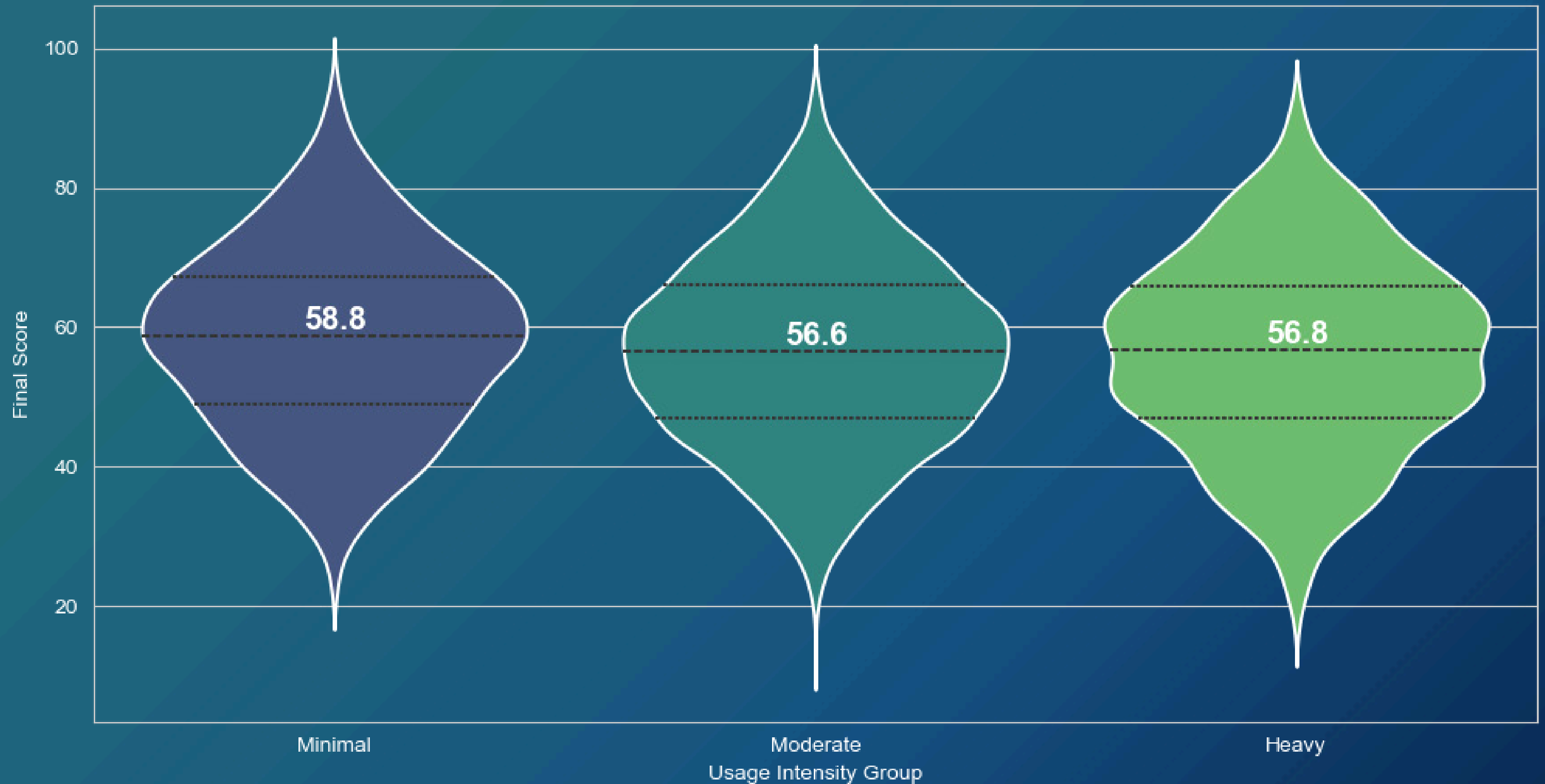


Distribution of Users

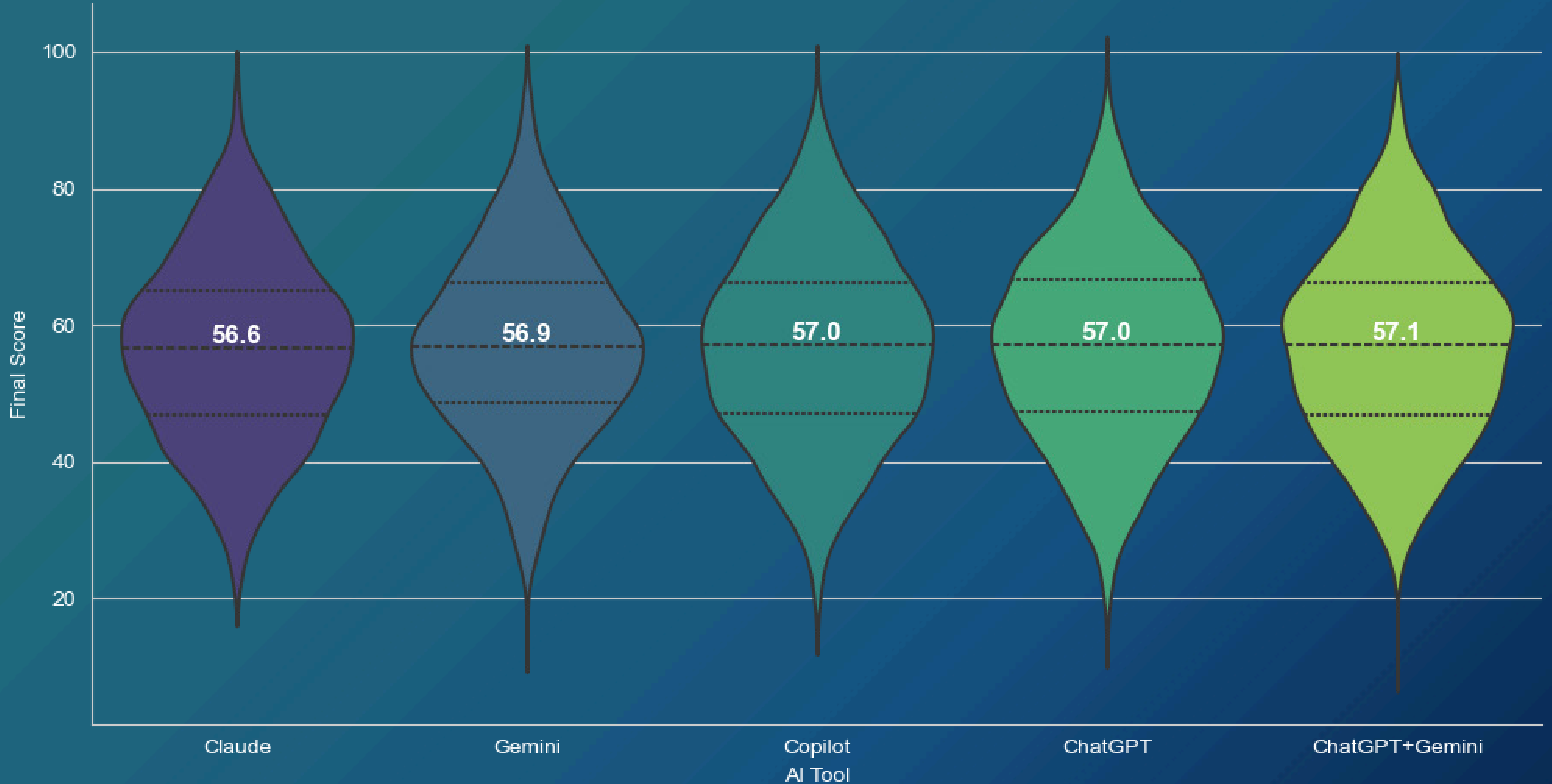
(Usage Intensity)



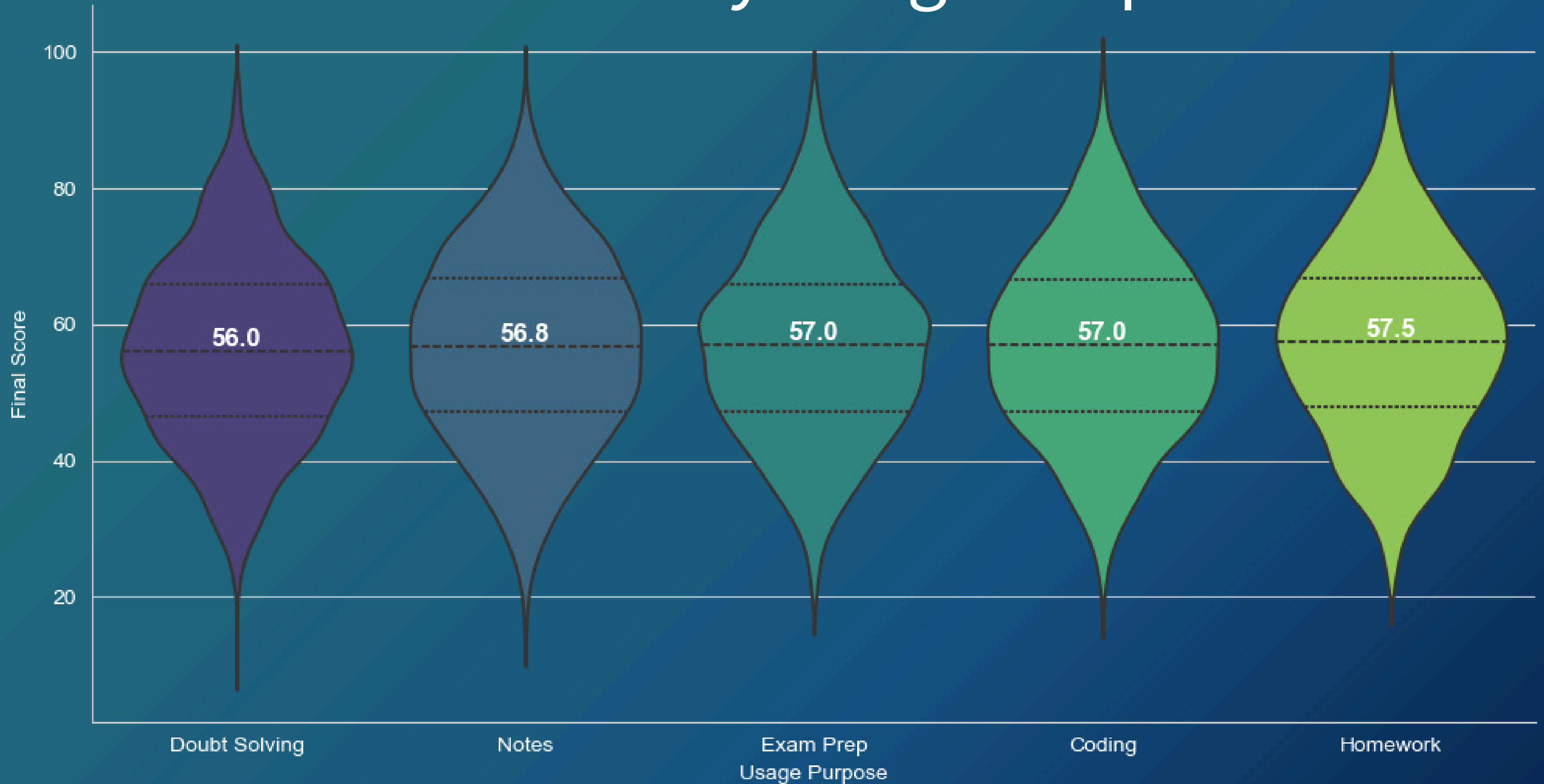
Final Score Distribution by Usage Intensity Group



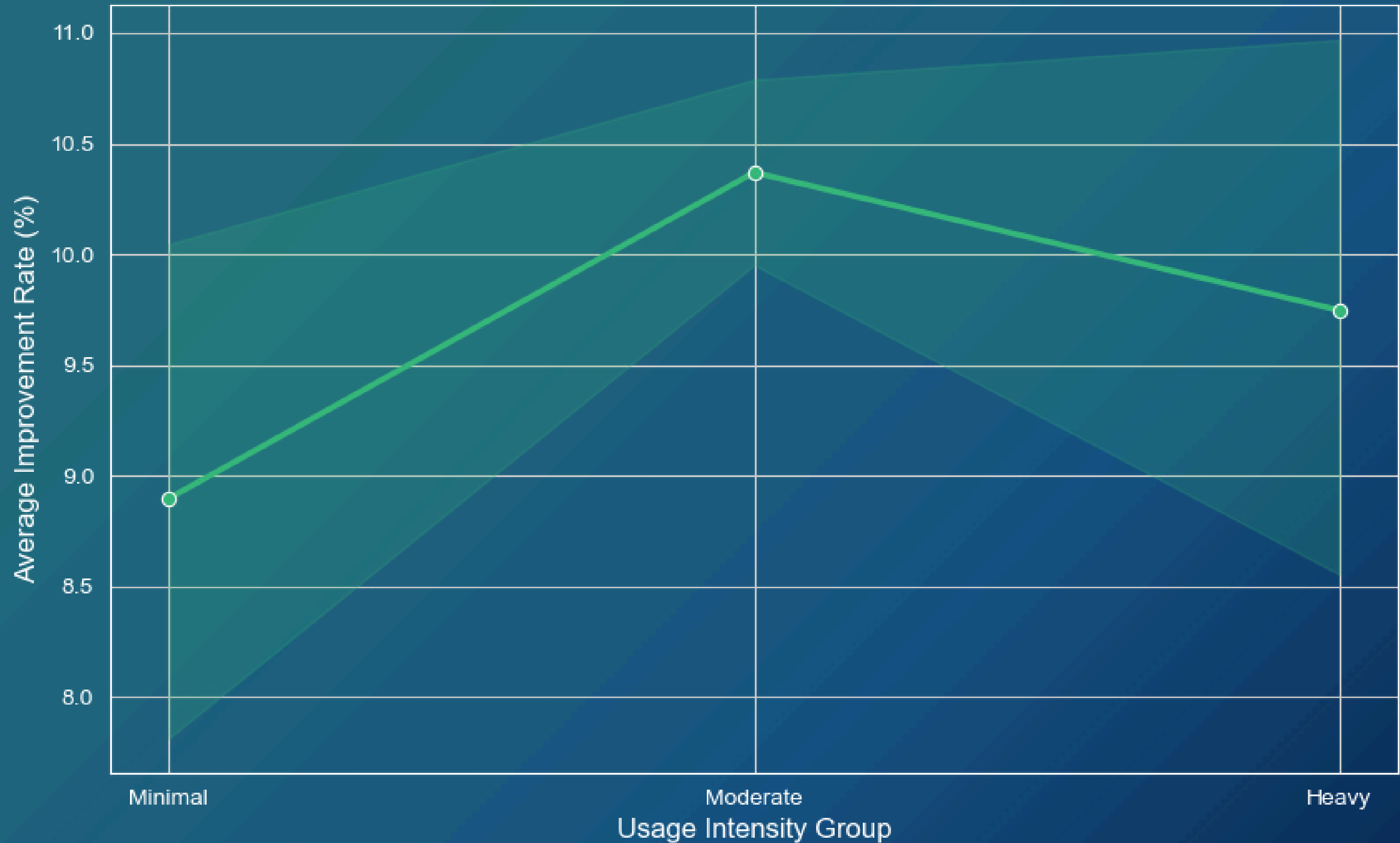
Final Score by AI Tool Used



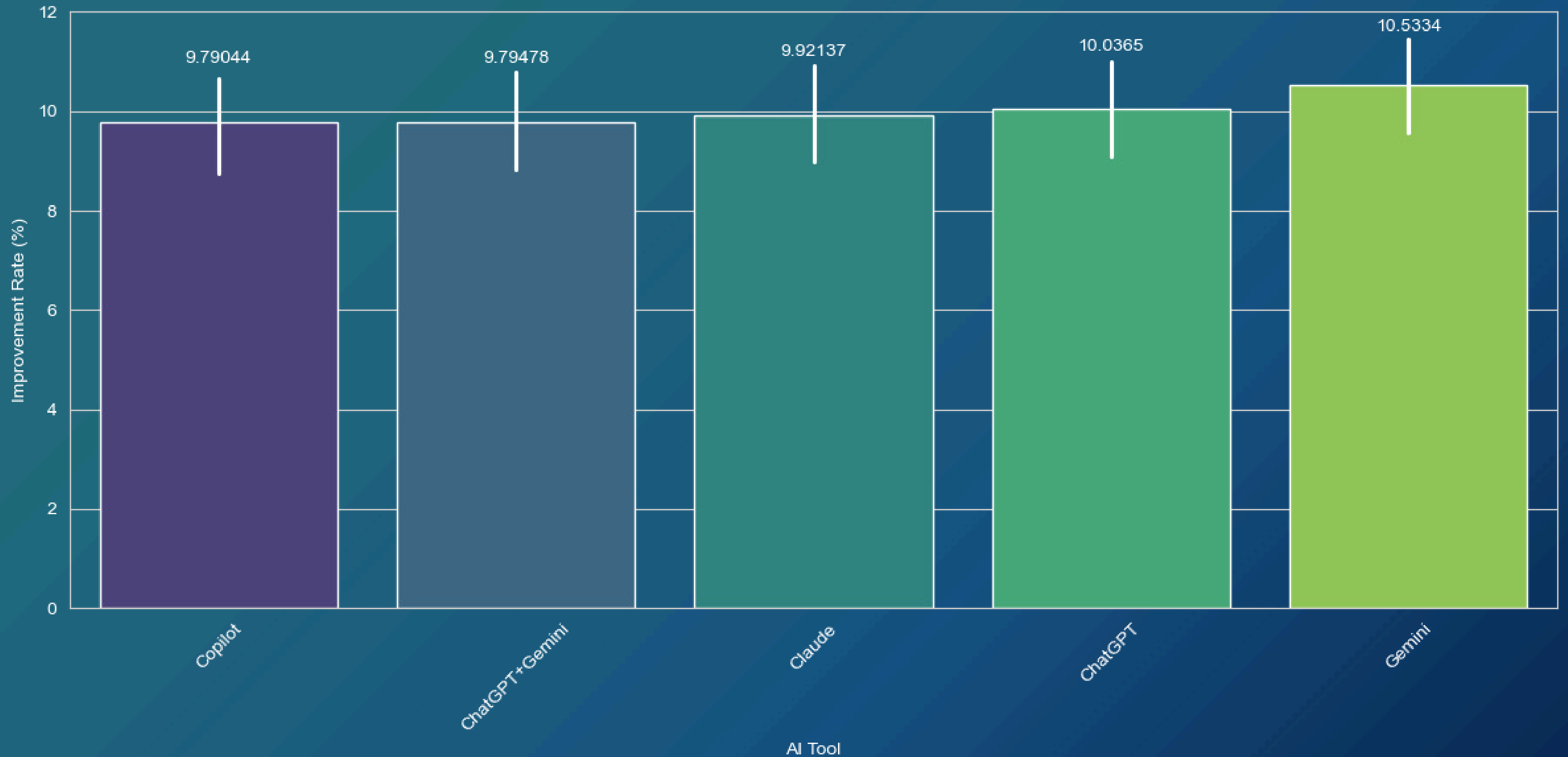
Final Score by Usage Purpose



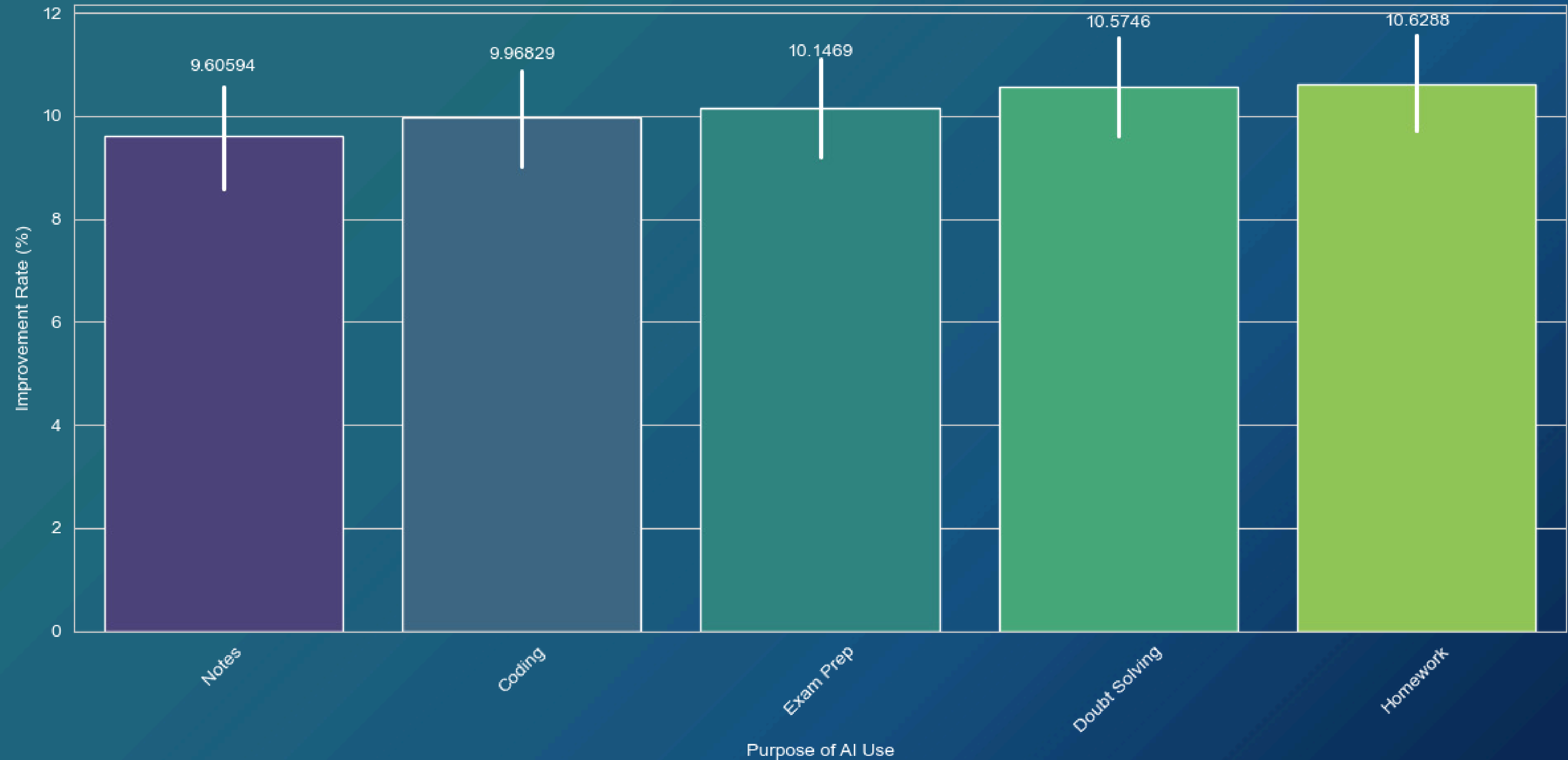
Improvement Rate by Usage Intensity Group



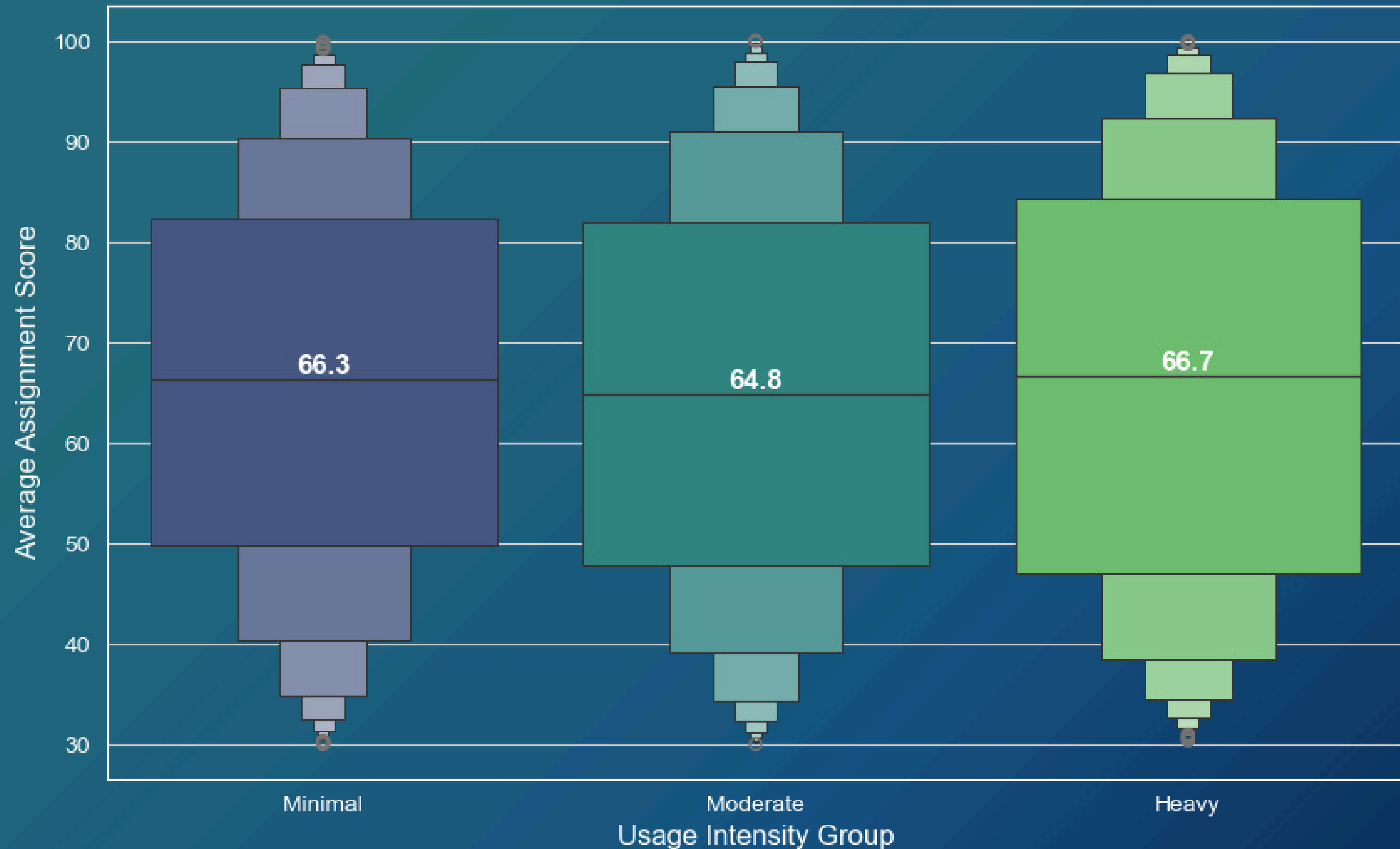
Improvement Rate by AI Tool Used



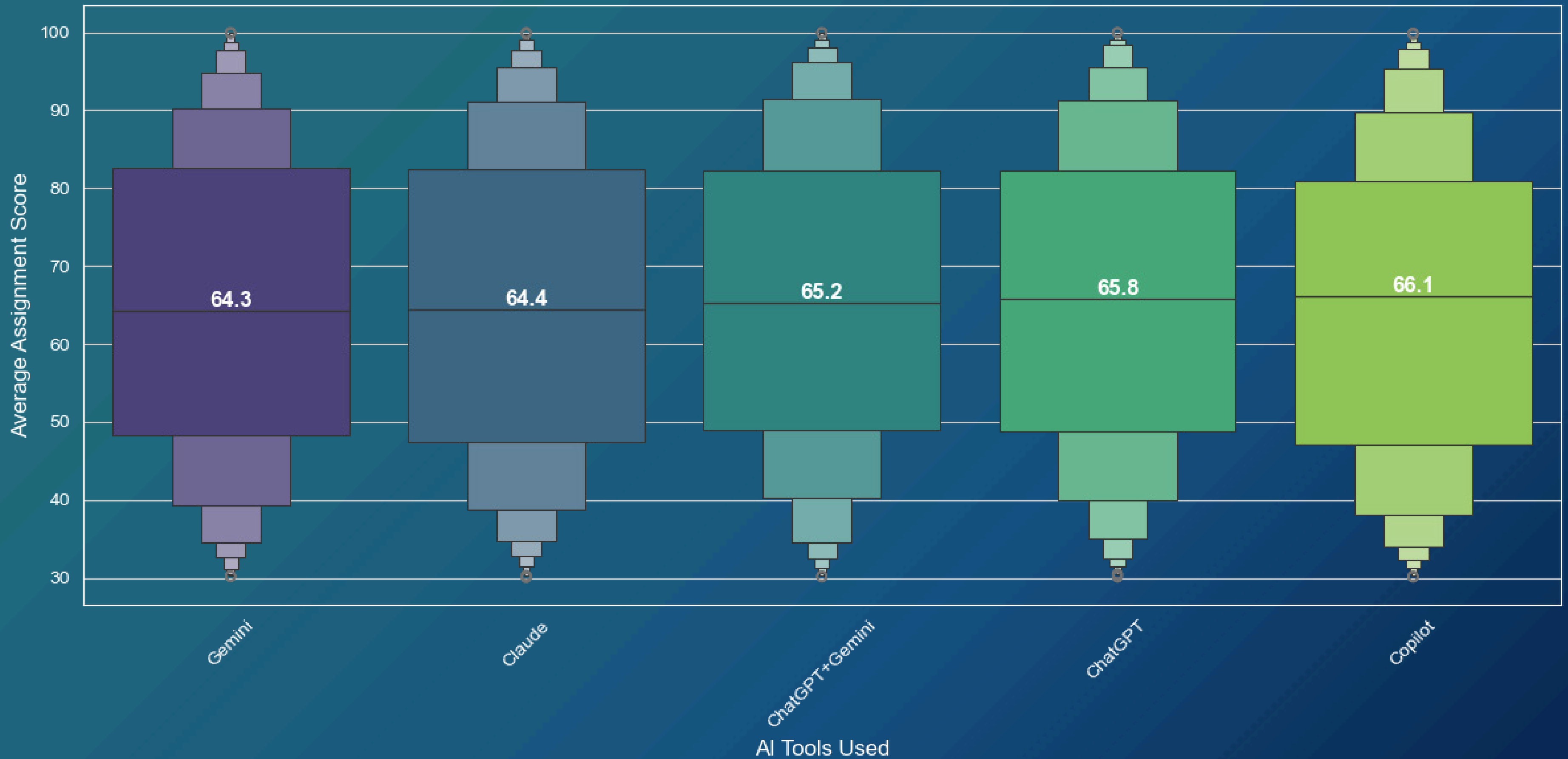
Improvement Rate by Usage Purpose



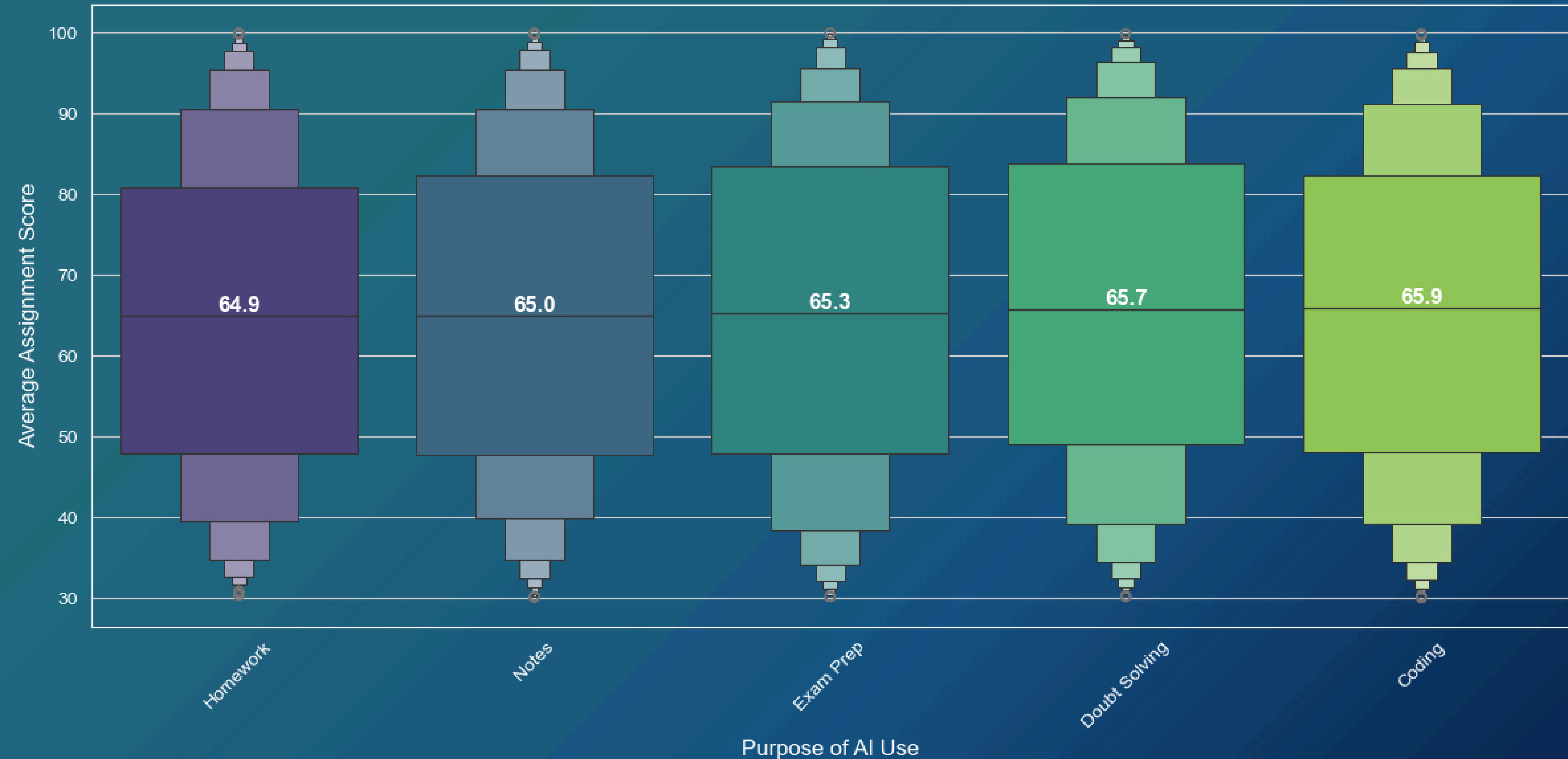
Average Assignment Score by Usage Intensity Group

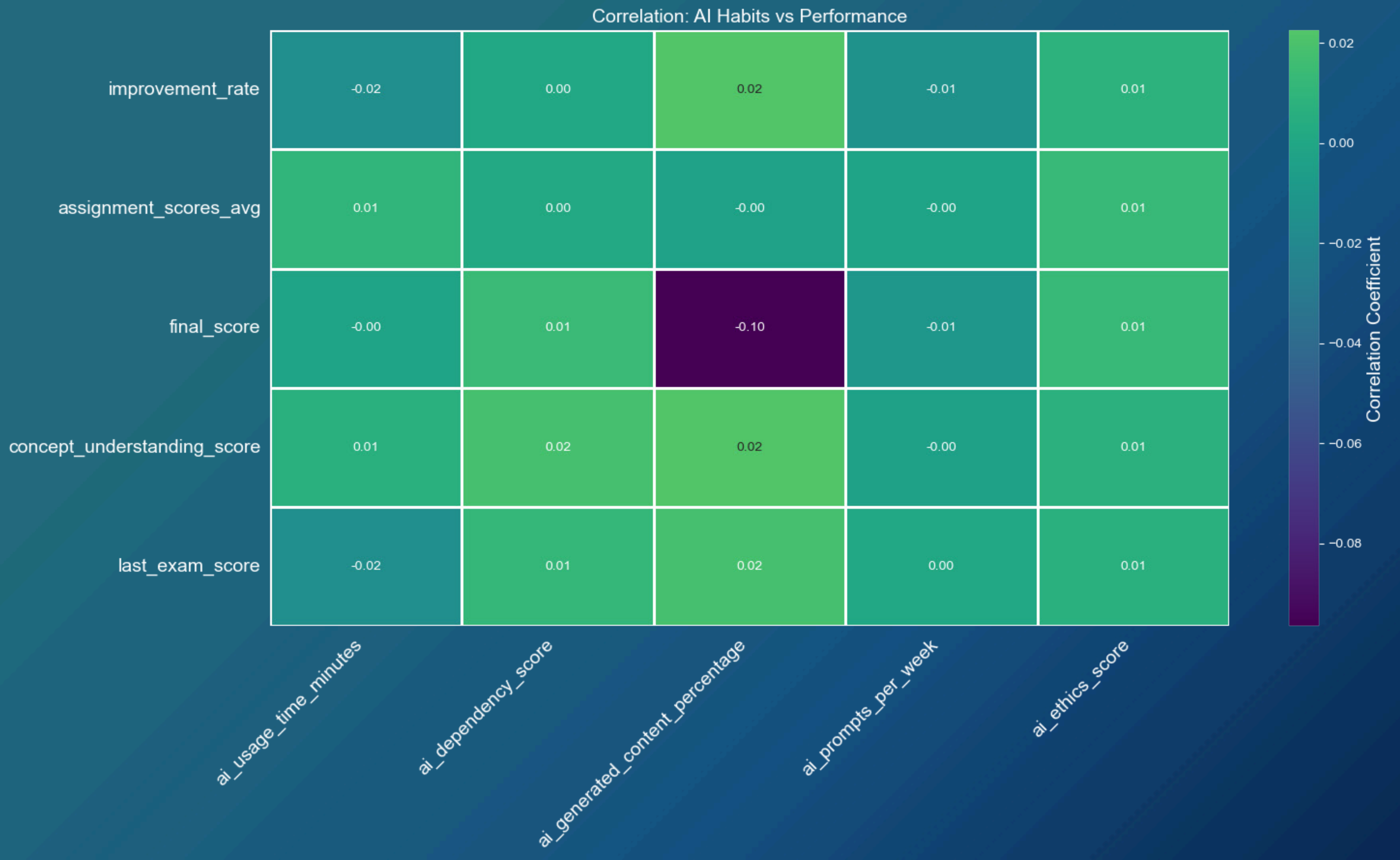


Average Assignment Score by AI Tool Used

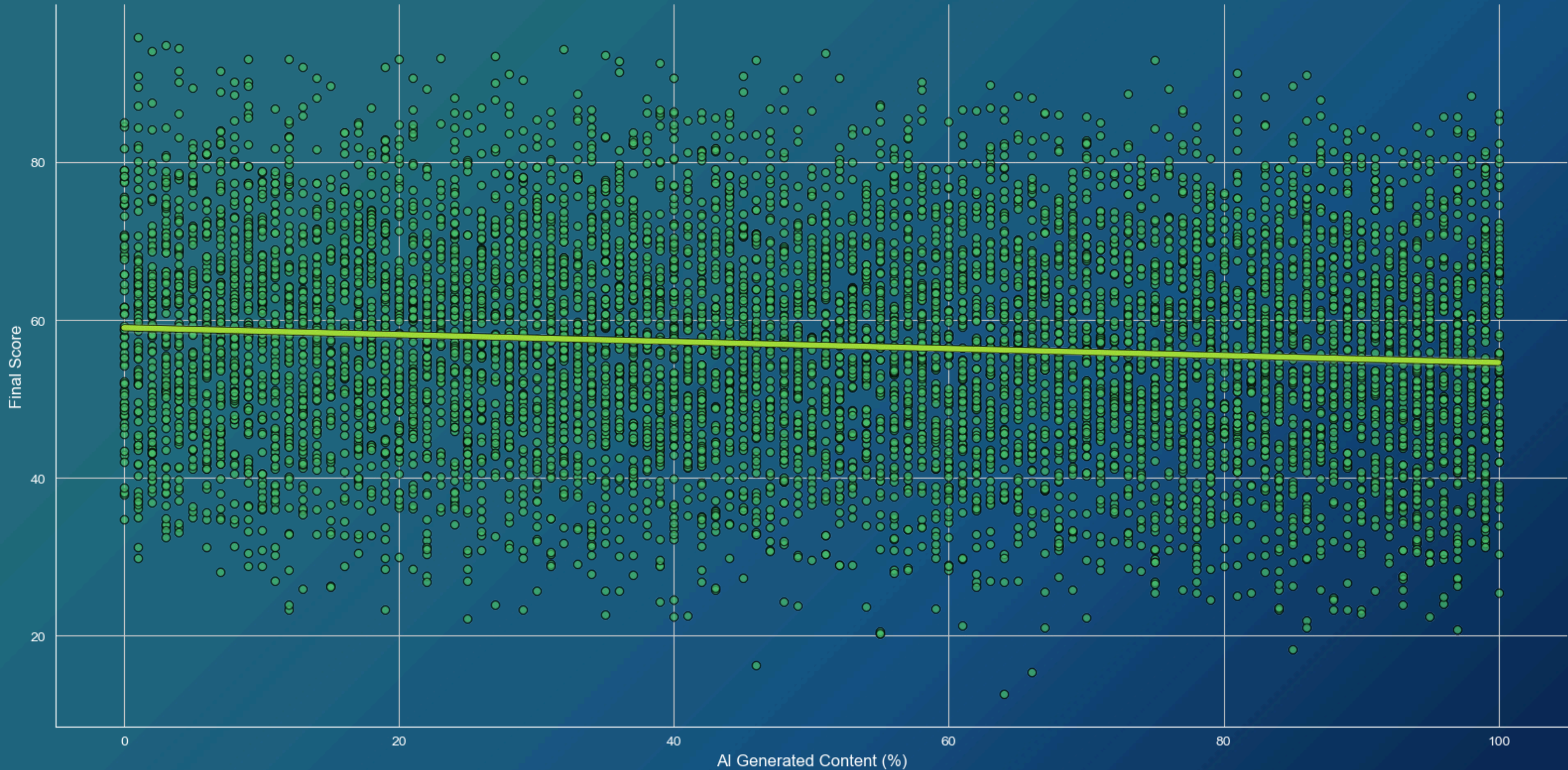


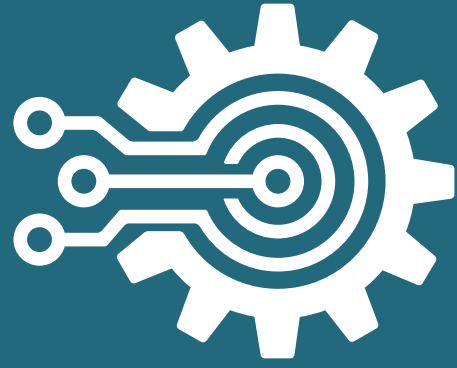
Average Assignment Score by Usage Purpose





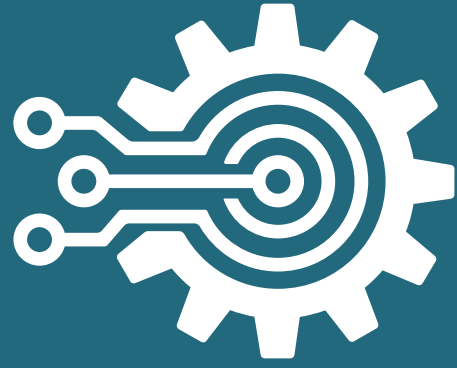
Final Score by AI Generated Content (%)





Key Findings:

- AI use can affect a student's academic outcomes in both positive and negative ways.
- Moderate use, focusing on homework, doubt solving, and exam prep, appears to be the best strategy for utilizing AI to help improve Academic gains.
- Heavy use of AI, specifically as it relates to over reliance of AI generated content, showed a sharp decline in not only final scores, but also improvement rates
- The tool used doesn't appear to matter as much as the intensity and reason of use



Conclusion

- Given the ubiquity of AI among students, AI regulations and policies should focus less on outright bans and focus more on regulating the ways in which AI can be used for academic work.
 - As the data shows, there are strategies of AI use that are conducive to positive academic growth, just as there are strategies which hinder it.
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